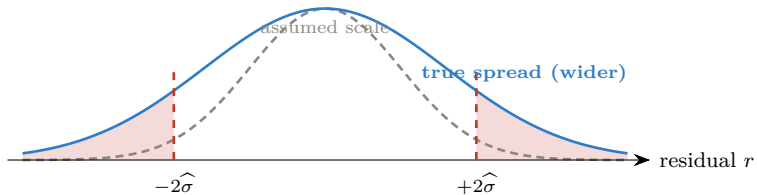
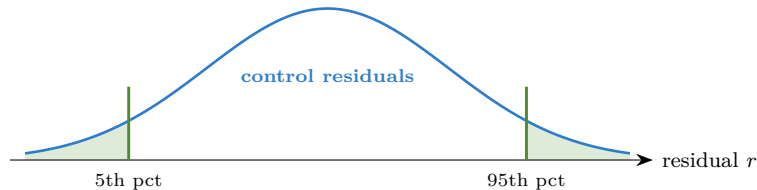


Fixed residual cutoff $|r| > 2\hat{\sigma}$
needs a trustworthy scale $\hat{\sigma}$



model understates spread $\Rightarrow$ cutoff excludes
 $\sim 21\%$ of *compatible* controls (target 5%)

Residual rank
cut at the actual controls' percentile



cutoff = empirical quantile of the actual controls
 $\Rightarrow$ excludes exactly the intended $\sim 10\%$, any scale